PREP 5 · LESSON 9

How much is useful?

Efficiency as a decimal and a percentage · Sankey diagrams

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Define power. DN 31 [1]

The rate at which energy is transferred (or work is done).

2 State what is meant by dissipated energy. DN 654 [1]

Energy transferred to the surroundings and stored in less useful ways.

3 Two motors lift the same load through the same height. Motor A is quicker. Which is more powerful, and why? DN 656 [1]

Motor A. It transfers the same energy in less time.

Key ideas LESSON 9

Show your working

4 A motor is supplied with 800 J. It transfers 600 J usefully. Use $\text{efficiency} = \frac{\text{useful output}}{\text{total input}}$

Calculate the efficiency of the motor, as a decimal and as a percentage. [3]

$\text{efficiency} = \frac{\text{useful output}}{\text{total input}}$ ✓

$\text{efficiency} = \frac{600}{800} = 0.75$ ✓

= 75% ✓

5 A 40 W lamp is 30% efficient. Calculate its useful power output. [2]

useful = efficiency × total ✓

useful = 0.30 × 40 = 12 W ✓

6 A lamp is supplied with 100 J and transfers 20 J by light. In a Sankey diagram, the wasted arrow shows [1]

80 J (20 J / 80 J / 120 J)

7 Explain why no device can have an efficiency greater than 1 (100%). [1]

Energy cannot be created, so the useful output cannot be more than the total input ✓